

Python Mini Project Case Studies

Case Study 1: Student Management System

Develop a menu-driven Student Management System using Python. The application should maintain student information in a JSON file.

Student Record Structure

```
{  
  "roll": 101,  
  "name": "Swarup",  
  "email": "swarup@gmail.com",  
  "course": "Python"  
}
```

Menu Options

1. Add Student
2. Display Students
3. Search Student By Roll
4. Exit

Requirements

Use a class named StudentManagement.

Implement methods: add_student(), display_students(), search_student().

Create a custom Context Manager named StudentDatabase.

Use a decorator named @logger to log operations.

Validate student name using Regular Expressions.

Validate email address using Regular Expressions.

Store records as a list of dictionaries.

Handle invalid input using Exception Handling.

Prevent duplicate roll numbers.